

FE-APP-02 - Outdoor Sales / Franchise App

Version: v1.0

Status: Draft for review

Depends on:

- FE-APP-00-Frontend_Architecture_and_Shared_UX_Foundation-v1.0
- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- DD_QA-01-End_to_End_Test_Strategy-v1.0
- DD_SALES-01-Lead_Territory_and_Franchise_Sales_Management-v1.0
- DD_FUL-02-Order_Capture-v1.0
- DD_ILM-CFG-01-Customer_Service-v1.0
- DD_RLM-CFG-01-HomePass_Configuration-v1.0
- DD_SIP-01-Package_Management-v1.0
- DD_BIL-01-PAY-01-Payment_Application-v1.0
- DD_EM-02-Contractor_and_Staff_Registry-v1.0
- DD_EM-02-Contractor_Team_Agent_Registry-v1.0
- FOUNDATION_AUTH
- FOUNDATION_FILE_STORAGE

1. Purpose

This DD defines the Outdoor Sales / Franchise App.

The app is used by door-to-door sales agents, franchise sales teams, and sales supervisors to capture prospects, check serviceability, select packages, upload KYC/consent evidence, convert qualified leads into FUL-02 orders, and track the order until handoff to fulfillment.

The app is Kenya/WIK-first but must remain deployable for other operators through configuration: language, currency, brand, feature flags, sales territories, KYC requirements, package availability, and payment instructions.

2. Design Position

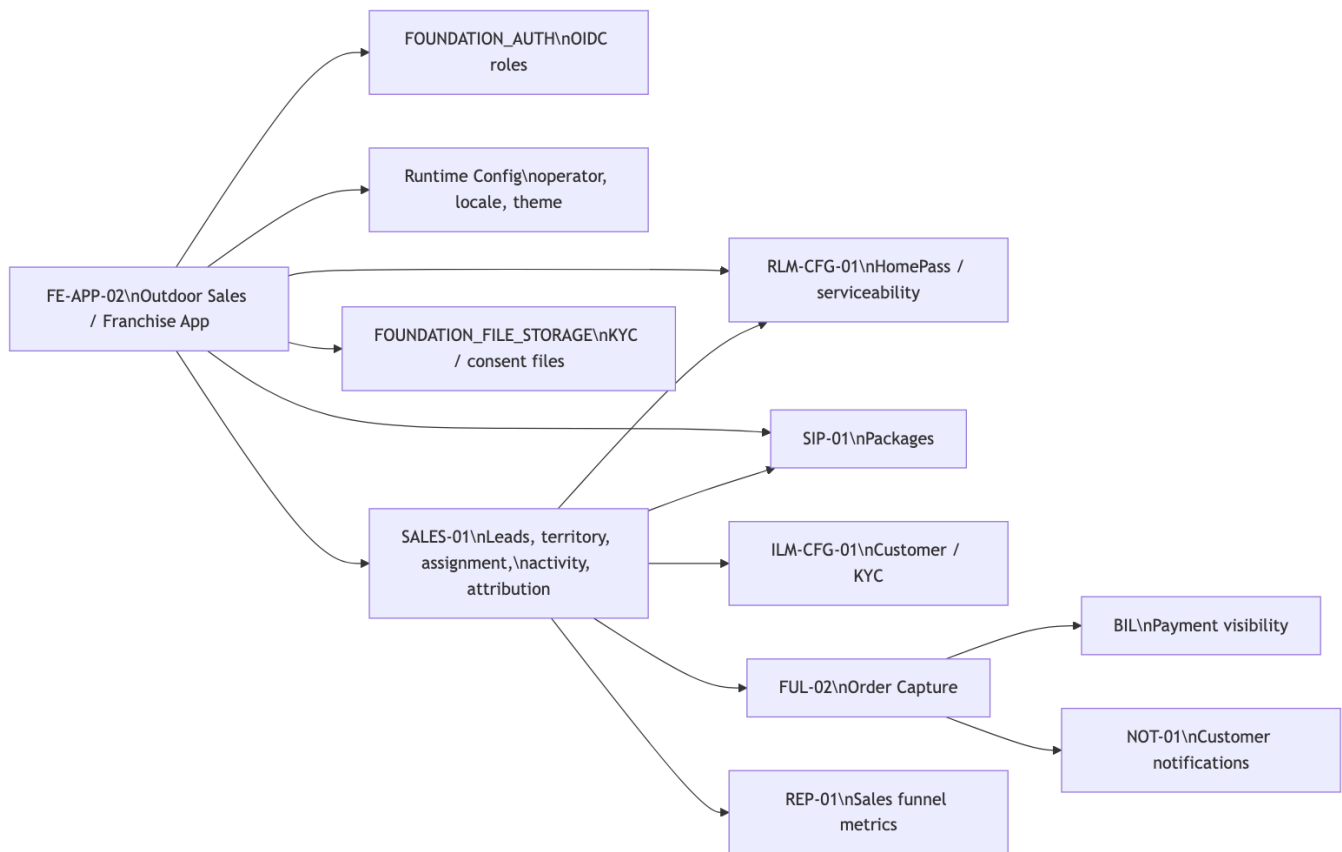
FE-APP-02 is a mobile-first operational sales app. It is not a generic CRM and it is not the order orchestration engine.

Ownership split:

Concern	Owner
Lead, territory, assignment, activity, attribution	SALES-01
Sales agent, team, franchise identity	EM-02
Customer/account/KYC source of truth after conversion	ILM-CFG-01
HomePass/serviceability	RLM-CFG-01
Package/catalog eligibility	SIP-01, PLM config where applicable
Submitted order lifecycle	FUL-02
Payment ledger/application	BIL-01-PAY-01
KYC/consent file storage	FOUNDATION_FILE_STORAGE
Customer communication	NOT-01 through backend-triggered flows

Illustration:

Figure 1 - Sales app module map



3. Application Scope

Included:

Area	Included features	Primary module APIs
Daily work	Assigned leads, follow-ups, territory list, route context.	SALES-01, EM-02, FOUNDATION_AUTH.
Lead capture	Prospect name, phone, location, HomePass, consent, package interest.	SALES-01, FOUNDATION_FILE_STORAGE.
Serviceability	HomePass search/detail, sellability warnings.	RLM-CFG-01.
Package selection	Eligible package list/detail, pricing display.	SIP-01, PLM package/service config APIs where needed.
KYC and consent	Document capture/upload, customer consent evidence.	FOUNDATION_FILE_STORAGE, ILM-CFG-01 after conversion, FUL-02 KYC step.
Lead qualification	Duplicate, eligibility, serviceability, package availability checks.	SALES-01; SALES calls ILM/RLM/SIP.
Convert to order	Submit qualified lead into order capture.	SALES-01, FUL-02.
Order tracking	View order phase after conversion.	FUL-02, BIL, W0, SUB-LM through read/status APIs where exposed.
Supervisor view	Team leads, conversion status, reassignment, follow-up aging.	SALES-01, EM-02, REP-01 optional summaries.
Offline drafts	Local lead/activity/KYC draft and retry queue.	Local encrypted storage; server authority remains SALES-01 and file APIs.

Non-scope:

Non-scope	Owner
Full Backoffice customer care	FE-APP-01.
Contractor installation execution	FE-APP-03, W0-01.
Customer self-service payment portal	FE-APP-04.
Commission payout calculation	Future finance/commission DD if required; SALES-01 emits attribution basis only.

Non-scope	Owner
Reporting portal dashboards	FE-APP-05, REP-01.

4. Users And Roles

Roles come from FOUNDATION_AUTH. Agent/team/franchise identity comes from EM-02.

Role family	Typical permissions	Module API link
Sales agent	View assigned leads, create lead, add activity, qualify, convert assigned lead.	FOUNDATION_AUTH role claims, SALES-01, EM-02.
Franchise sales agent	Same as sales agent but scoped to franchise/territory.	FOUNDATION_AUTH, EM-02, SALES-01.
Sales supervisor	View team leads, assign/reassign, resolve duplicates, monitor follow-ups.	SALES-01, EM-02.
Franchise manager	View franchise leads and agent work, limited reassignment.	SALES-01, EM-02, optional REP-01.
KYC reviewer	Not primary app user; reviews in Backoffice.	FE-APP-01, ILM, FUL-02.
Read-only auditor	View selected sales records only.	SALES-01, REP-01.

Implementation rule: the app can hide navigation/actions from role claims, but all APIs must still enforce backend authorization.

5. Navigation Model

Recommended mobile navigation:

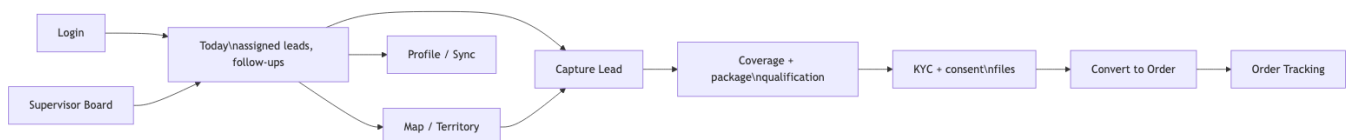
Tab	Purpose	Module API link
Today	Assigned leads, follow-ups, pending sync, quick capture.	GET /api/sales/agents/{agent_id}/daily-work.
Map	Territory view and nearby HomePass/prospects.	SALES-01 territories, RLM-CFG-01 HomePass APIs.
Capture	New lead, KYC/consent, package interest.	POST /api/sales/leads, file upload APIs.
Orders	Converted leads and FUL order tracking.	SALES-01 conversion, FUL-02 order status APIs.
Profile	Agent profile, operator context, sync status, logout.	FOUNDATION_AUTH, EM-02, runtime config.

Supervisor users may see an additional Team tab:

Tab	Purpose	Module API link
Team	Team lead board, assignment, aging, performance counters.	SALES-01, EM-02, optional REP-01.

Illustration:

Figure 2 - Sales app screen flow



6. Application Shell

Required shell components:

Component	Behavior	Module/config link
Operator context	Shows operator/country/brand.	Runtime config from FE-APP-00.
Offline indicator	Shows online/offline/sync pending.	Local sync manager.
Sync queue badge	Shows unsynced leads/files/activities.	Local queue plus SALES/file API responses.
Bottom navigation	Today, Map, Capture, Orders, Profile.	Frontend route config.

Component	Behavior	Module/config link
Quick capture	Always available for field work.	Local draft first, then SALES-01.
Language selector	If enabled for operator.	Runtime config/i18n.
Profile	User, roles, franchise/team, logout.	FOUNDATION_AUTH, EM-02.

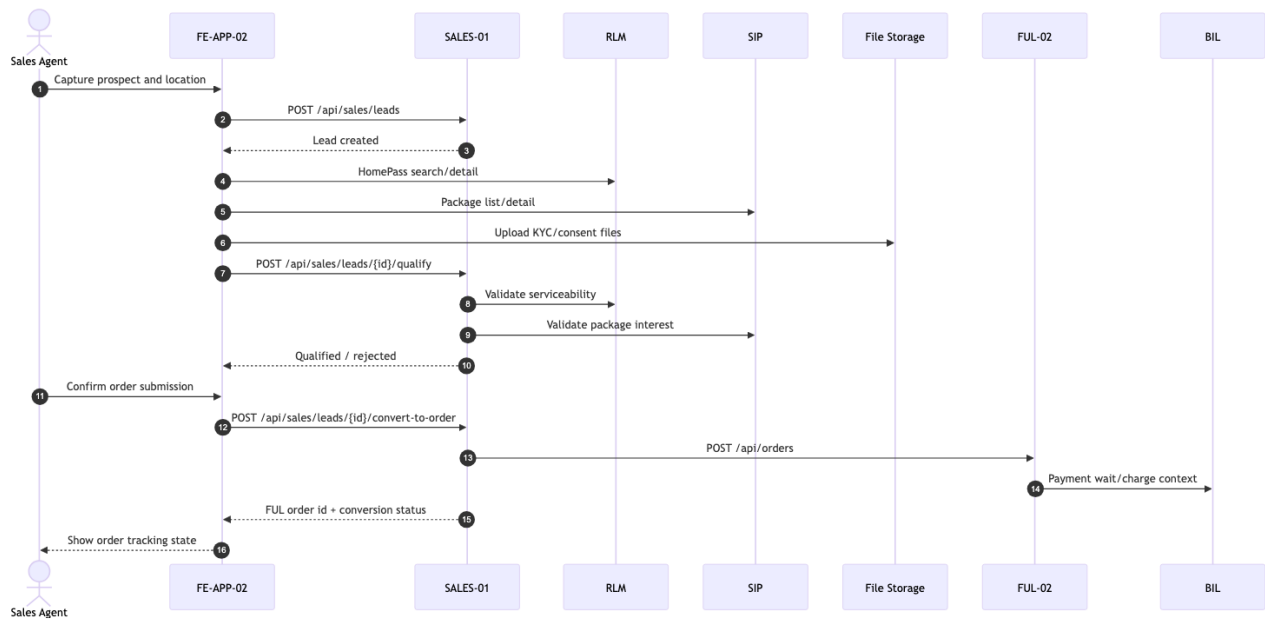
7. Main User Flows

7.1 Lead Capture To Order

This is the main happy path.

Illustration:

Figure 3 - Lead to order flow



Flow:

Step	App behavior	Module API link
1. Start capture	Agent enters prospect name/phone/location and consent.	Local draft, then POST /api/sales/leads.
2. Check HomePass	Agent searches/selects HomePass or captures location for later matching.	RLM - CFG-01 HomePass search/detail APIs.
3. Select package	App lists packages eligible for operator/region/HomePass.	SIP-01, PLM service/package APIs.
4. Upload evidence	App captures KYC/consent photos/files.	FOUNDATION_FILE_STORAGE.
5. Qualify	App asks SALES-01 to validate duplicate, serviceability, package, customer eligibility.	POST /api/sales/leads/{lead_id}/qualify.
6. Convert	Agent confirms order.	POST /api/sales/leads/{lead_id}/convert-to-order.
7. Track	App shows FUL order status and next action.	FUL-02 order status APIs; BIL/WO status via FUL or owner APIs where exposed.

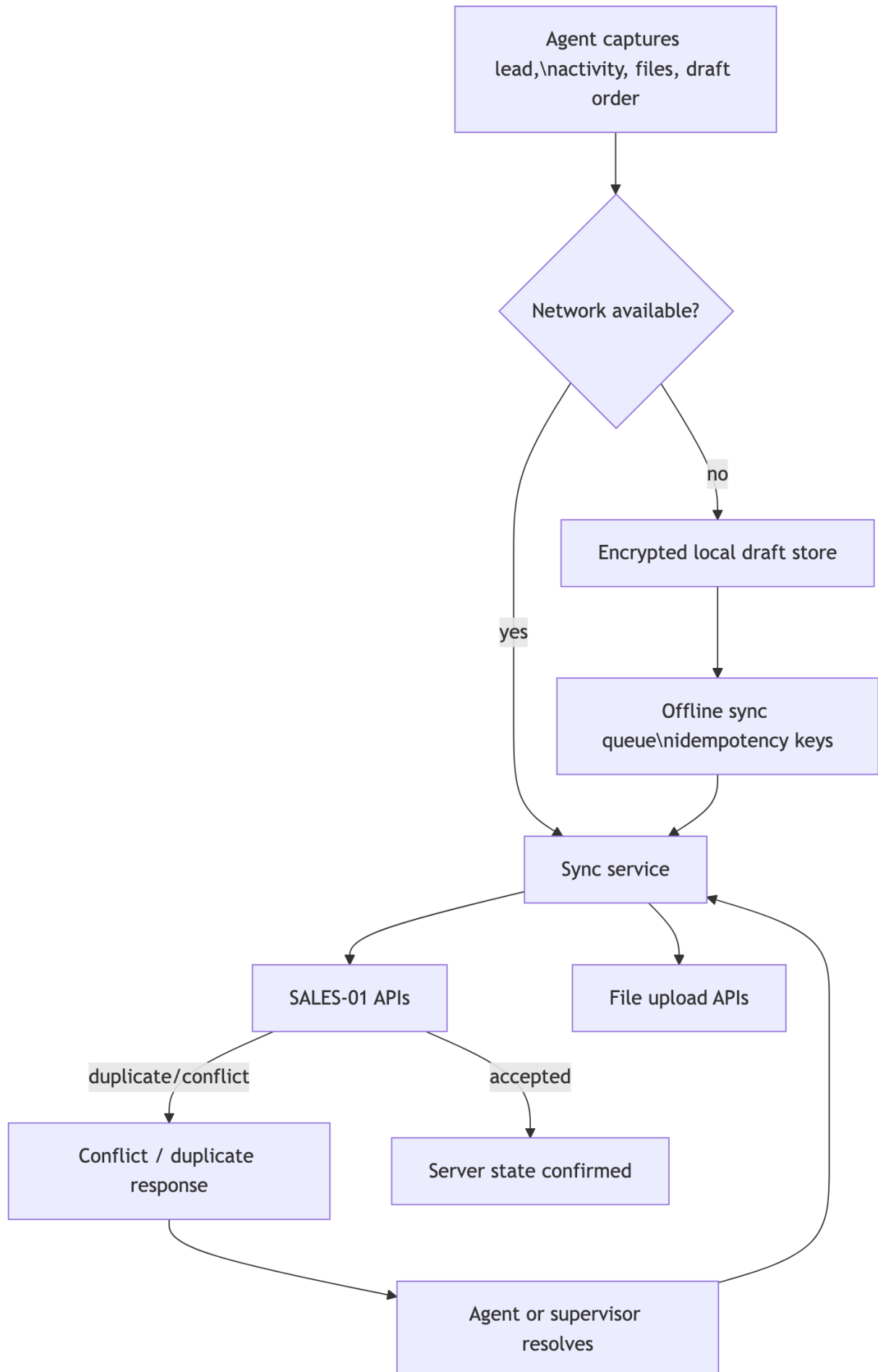
The app must show that the lead is not an order until conversion succeeds.

7.2 Offline Lead Draft

Offline mode allows field capture without network. It does not create an authoritative server lead until sync succeeds.

Illustration:

Figure 4 - Offline sync



Rules:

- Store drafts in encrypted local storage where platform supports it.
- Generate stable local draft ids and idempotency keys.
- Queue files separately from lead JSON where needed.
- Sync files first if the lead payload references uploaded file refs.
- When server returns duplicate/conflict, show resolution screen instead of creating another local lead.
- Do not confirm order conversion while offline. Conversion requires server connectivity.

7.3 Supervisor Reassignment

Supervisors can reassign leads when an agent is unavailable or territory ownership changes.

Step	App behavior	Module API link
List team leads	Filter by team, territory, status, age.	GET /api/sales/leads?... , EM team APIs.
Select lead	Open lead summary and activity history.	GET /api/sales/leads/{lead_id}.
Reassign	Choose target agent/team.	EM agent/team lookup APIs, POST /api/sales/leads/{lead_id}/assign.
Notify	Backend emits event; ICN/NOT handle notifications where configured.	SalesLeadAssigned event, ICN/NOT.

8. Screen Groups

8.1 Today

Screen/action	Purpose	Module API link	Representative API
Daily work list	Show assigned leads, due follow-ups, pending conversions.	SALES-01.	GET /api/sales/agents/{agent_id}/daily-work.
Lead quick action	Call, navigate, open capture, mark follow-up.	SALES-01.	POST /api/sales/leads/{lead_id}/activities.
Sync warning	Show pending offline items.	Local sync queue.	Local only; server sync uses SALES/file APIs.
Team badge	Show agent team/franchise.	EM-02.	EM agent/team detail APIs.

8.2 Map And Territory

Screen/action	Purpose	Module API link	Representative API
Territory list	Show assigned sales areas.	SALES-01, EM-02.	GET /api/sales/territories?...
Nearby HomePass	Show nearby serviceable premises.	RLM-CFG-01.	HomePass search/detail APIs.
Lead pins	Show assigned leads on map.	SALES-01.	GET /api/sales/leads?territoryId=...
Capture from map	Start lead with location/HomePass prefilled.	SALES-01, RLM-CFG-01.	Local draft then POST /api/sales/leads.

8.3 Lead Capture

Screen/action	Purpose	Module API link	Representative API
Create lead	Create prospect record.	SALES-01.	POST /api/sales/leads.
Edit lead	Update allowed pre-conversion fields.	SALES-01.	PATCH /api/sales/leads/{lead_id}.
Add package interest	Record candidate package(s).	SALES-01, SIP-01.	PATCH /api/sales/leads/{lead_id} after package lookup.
Capture consent	Attach consent evidence.	FOUNDATION_FILE_STORAGE, SALES-01.	Upload file, then include file ref on lead/activity.
Add visit note	Record outcome/follow-up.	SALES-01.	POST /api/sales/leads/{lead_id}/activities.

8.4 Coverage And Package Selection

Screen/action	Purpose	Module API link	Representative API
HomePass lookup	Validate serviceability.	RLM - CFG - 01.	GET /api/homepasses/{homepass_id}.
Package list	Show eligible packages.	SIP - 01.	Package list/detail APIs.
Service detail	Explain service/package components.	PLM - CFG - 01, SIP - 01.	GET /api/services/by-code/{service_code}, GET /api/packages/{package_id}.
Qualification	Validate duplicate, package, HomePass, customer eligibility.	SALES - 01.	POST /api/sales/leads/{lead_id}/qualify.

8.5 KYC And Consent

Screen/action	Purpose	Module API link	Representative API
Capture document	Take photo/upload file.	FOUNDATION_FILE_STORAGE.	Authorized upload/init APIs.
Attach to lead	Link file refs to lead draft/conversion request.	SALES - 01.	PATCH /api/sales/leads/{lead_id} or conversion payload.
Customer/KYC creation	Create customer/order KYC context after conversion.	SALES - 01, FUL - 02, ILM - CFG - 01.	POST /api/sales/leads/{lead_id}/convert-to-order.
KYC review status	Show waiting/rejected/approved after order submission.	FUL - 02, ILM - CFG - 01.	FUL order status/user-task state APIs.

8.6 Convert To Order

Screen/action	Purpose	Module API link	Representative API
Review summary	Show customer, HomePass, package, price, consent, attribution.	SALES/RLM/SIP read APIs.	Lead detail, HomePass detail, package detail.
Submit order	Convert qualified lead into FUL order.	SALES - 01, FUL - 02.	POST /api/sales/leads/{lead_id}/convert-to-order; SALES calls POST /api/orders.
Conversion result	Show order id/status.	SALES - 01, FUL - 02.	Lead conversion detail and FUL order detail APIs.
Retry conversion	Safe retry only with same idempotency key.	SALES - 01.	Same convert endpoint with same Idempotency-Key.

8.7 Orders

Screen/action	Purpose	Module API link	Representative API
Converted leads	Show leads with order ids.	SALES - 01.	GET /api/sales/leads?status=CONVERTED.
Order status	Track order phase.	FUL - 02.	FUL order detail/status APIs.
Payment wait	Show pending/received payment status.	FUL - 02, BIL - 01 - PAY - 01.	FUL order status; BIL payment status where exposed.
Install wait	Show install/WO status.	FUL - 02, WO - 01.	FUL order status; WO detail if linked and authorized.
Activation status	Show activation final state.	FUL - 03, SUB - LM.	FUL/SUB status APIs where exposed.

8.8 Supervisor Team Board

Screen/action	Purpose	Module API link	Representative API
Team leads	List by team/franchise/status/age.	SALES - 01, EM - 02.	GET /api/sales/leads?...
Reassign	Move lead to another agent/team.	SALES - 01, EM - 02.	POST /api/sales/leads/{lead_id}/assign.
Duplicate resolution	Review duplicate warning and close/reopen.	SALES - 01, ILM - CFG - 01.	SALES duplicate handling APIs; ILM customer search.

Screen/action	Purpose	Module API link	Representative API
Performance snapshot	Show daily conversion/follow-up counters.	SALES-01, optional REP-01.	SALES summary APIs or REP dashboard APIs.

9. API Integration Details

9.1 Required Headers

Header	When	Purpose
Authorization	All protected APIs.	JWT from FOUNDATION_AUTH.
X-Correlation-Id	All important reads and commands.	Trace across SALES/FUL/BIL/WO.
Idempotency-Key	Create lead, upload finalization, convert order.	Safe retry from weak networks.
X-Operator-Code	Where endpoint requires explicit operator.	Operator scoping.
Accept-Language	All frontend calls.	Localized errors/messages.

9.2 Lead Create Request

```
POST /api/sales/leads
Authorization: Bearer <jwt>
Idempotency-Key: sales-lead-offline-draft-8f60c
Content-Type: application/json

{
  "operatorCode": "WIK",
  "sourceChannel": "D00R_TO_D00R",
  "prospectName": "Jane Njeri",
  "primaryPhone": "+254712345678",
  "homepassId": "hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P",
  "territoryCode": "SALES-KE-NRB-LAVINGTON",
  "geo": { "lat": -1.2831, "lng": 36.7820 },
  "consentCaptured": true,
  "packageInterest": [
    { "packageId": "pkg_INET_50M_HOME", "priority": 1 }
  ]
}
```

9.3 Convert To Order Request

```
POST /api/sales/leads/lead_01J9W7Y0Z3/convert-to-order
Authorization: Bearer <jwt>
Idempotency-Key: convert-lead_01J9W7Y0Z3
Content-Type: application/json

{
  "selectedPackageId": "pkg_INET_50M_HOME",
  "customerDraft": {
    "fullName": "Jane Njeri",
    "primaryPhone": "+254712345678",
    "idDocumentType": "NATIONAL_ID",
    "idDocumentNumber": "12345678"
  },
  "kycFileRefs": ["file_kyc_front_01", "file_kyc_back_01"],
  "salesAttribution": {
    "leadId": "lead_01J9W7Y0Z3",
    "agentId": "agt_01HZ_SALES_04",
    "teamId": "team_sales_nrb_04",
    "franchiseContractorId": "contractor_ke_nairobi_a",
    "territoryCode": "SALES-KE-NRB-LAVINGTON"
  }
}
```

10. Offline And Sync Rules

Allowed offline:

Item	Offline allowed?	Notes
New lead draft	Yes	Stored locally until synced to SALES-01.
Lead activity	Yes	Queue with idempotency key.
KYC/consent file capture	Yes	Store encrypted local file until upload succeeds.

Item	Offline allowed?	Notes
HomePass/package cached lookup	Limited	Cache recent lookups with freshness marker.
Lead qualification	No	Must call SALES-01 because duplicate and eligibility require server state.
Convert to order	No	Must be online.
Payment confirmation	No	Must come from BIL/FUL status, not local claim.

Conflict handling:

Conflict	App behavior
Duplicate lead	Show existing lead if user is authorized; offer follow-up or supervisor escalation.
Existing customer	Show warning and route to allowed customer/order path.
HomePass no longer sellable	Block conversion and keep lead as follow-up/lost.
Package no longer active	Ask agent to select another package.
File upload failed	Keep draft pending; do not convert until required files are uploaded.

11. Regional And Operator Configuration

Runtime config must control:

Config	Example
Operator/country	WIK, KE.
Currency/locale/timezone	KES, en-KE, Africa/Nairobi.
Brand/theme	Logo, app name, colors.
KYC requirements	Required document types by operator.
Consent text	Operator legal terms and language.
Payment instruction text	M-Pesa/paybill/channel guidance where configured.
Feature flags	leadMapEnabled, offlineKycEnabled, supervisorBoardEnabled, commissionPreviewEnabled.
Package display rules	Show/hide packages by operator/region/channel.

No operator-specific labels or legal text should be hardcoded.

12. Security And Privacy

Rules:

- Store only necessary draft data locally.
- Encrypt local drafts/files where supported.
- Clear local data on logout, account switch, or app wipe policy.
- Do not expose customer ID documents in logs.
- Do not use public object URLs for KYC/consent files.
- Do not allow one franchise to see another franchise's leads unless role/config allows it.
- Show customer phone/ID details only according to role and masking rules.
- Never treat offline state as final server state.

13. Observability

Track:

Metric/event	Purpose
Lead create success/failure	Sales app reliability.
Offline draft count and age	Detect stuck field work.
Sync failure reason	Support weak-network operations.

Metric/event	Purpose
Conversion success/failure	Sales funnel health.
File upload failure	KYC evidence reliability.
Duplicate conflict count	Data quality and territory overlap.
API latency by module	Identify slow HomePass/package/order calls.

Frontend logs must include correlation id and local draft id, but not raw KYC images or sensitive ID numbers.

14. E2E Scenarios

Required from DD_QA-01:

Scenario	Expected result
New WIK customer activation	Lead converted, order created, payment/KYC/install/activation complete.
Offline lead draft sync	Draft survives offline capture, syncs once, idempotent retry safe.
Duplicate lead conflict	Duplicate warning shown, no duplicate server lead created.
Package/HomePass not eligible	Conversion blocked with module error mapped to user message.
Supervisor reassignment	Lead owner changes, old owner cannot mutate lead, new owner can continue.

15. Implementation Notes

Recommended build order:

1. Implement app shell, auth, runtime config, operator branding.
2. Implement local draft and sync queue foundation.
3. Implement Today list from SALES daily work API.
4. Implement create lead online and offline.
5. Implement HomePass/package lookup screens.
6. Implement KYC/consent file capture and upload.
7. Implement qualify and convert-to-order flow.
8. Implement order tracking.
9. Implement supervisor board and reassignment.
10. Add E2E tests and offline sync tests.

Do not start with order submission before lead ownership and offline draft behavior are implemented. Otherwise the app will push incomplete prospects into FUL-02.

16. Acceptance Checklist

- [] Every screen/action names its module API owner.
- [] SALES-01 is used for leads/territory/assignment/activity/attribution.
- [] FUL-02 is used only after lead conversion to order.
- [] Offline drafts are not treated as server leads until synced.
- [] Conversion requires online state and Idempotency-Key.
- [] File uploads use FOUNDATION_FILE_STORAGE.
- [] Roles come from FOUNDATION_AUTH; agent/team/franchise identity comes from EM.
- [] Kenya/WIK labels, legal text, currency, and language are configurable.
- [] E2E tests cover lead capture to activation and offline draft sync.